list
cea tech

Smart Digital Systems

FROM RESEARCH TO INDUSTRY

INSTITUT CARNOT CEA LIST
université PARIS-SACLAY
UGA Université Grenoble Alpes

Who are we?

The Alternative Energies and Atomic Energy Commission (CEA) is a French public Research and Technology Organisation (RTO and Fundamental science), with the following main fields of expertise:

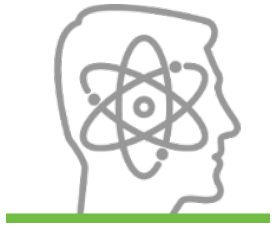
Low carbon energies

Digital transition

Health

Defense and security

Circular economy



Fundamental research

20,181
Employees

~ 5,000
Publications per year

~ 700
Industrial partnerships

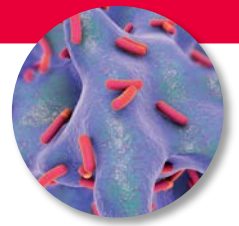
CEA'S missions

1 Defense and national security

2 Energy independence

3 Economic competitiveness of industry

DRF
Fundamental research



DAM
Security defense

DES
Civil nuclear & new energies

DRT
Technology research

1000 pers.

leti

[Micronano-technologies]

liten

[New energy technologies]

list

[Smart digital technologies]

Our work and expertise in Digital

Computing & components

650 researchers

High-performance embedded and **quantum computing**

Microelectronic fields

Low-power technologies

Specialised architectures, adaptation of **algorithms and code**

Artificial intelligence

500 researchers

From **data and algorithms** to dedicated **components**

Solutions adapted for **embedded, trust and sobriety**

Vision and image analysis

Automatic language processing and semantic analysis

System integration

600 researchers

From **sensors** to **human-machine interfaces** (HMI)

Instrumentation and signal processing chain

Extended reality

Software engineering and formal methods for **design and validation**

Digital twinning and **simulation**

Security communications

400 researchers

Cybersecurity

Quantum communication

Digital **infrastructure** (telecom, networks)

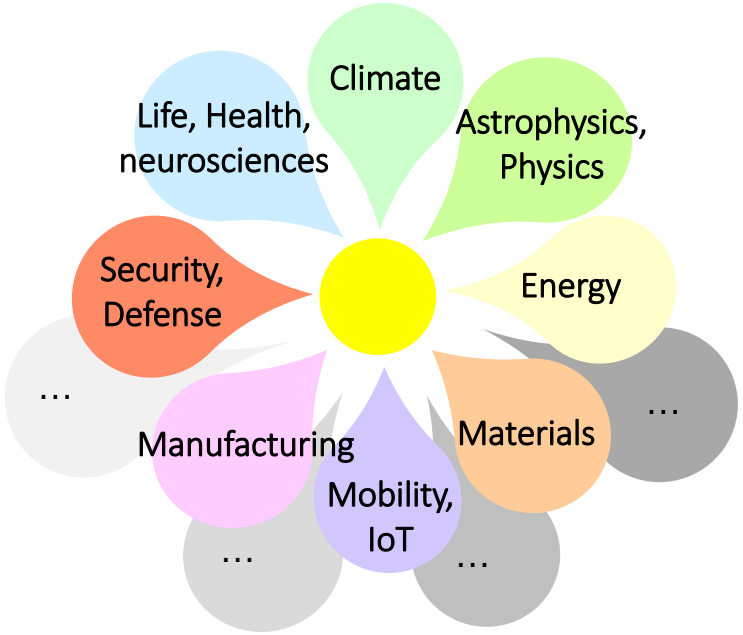
Blockchain for resilient distributed systems

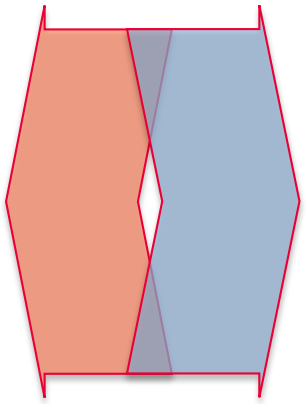
For **key industrial application** domains:

- **Industry of the futur**, robotics: man-robot collaboration, autonomy, smart diagnosis, system optimisation
- **Energy**: conversion and transport management, smart energy networks, power electronics
- **Materials**: digital design, process, 3D printing and additive manufacturing

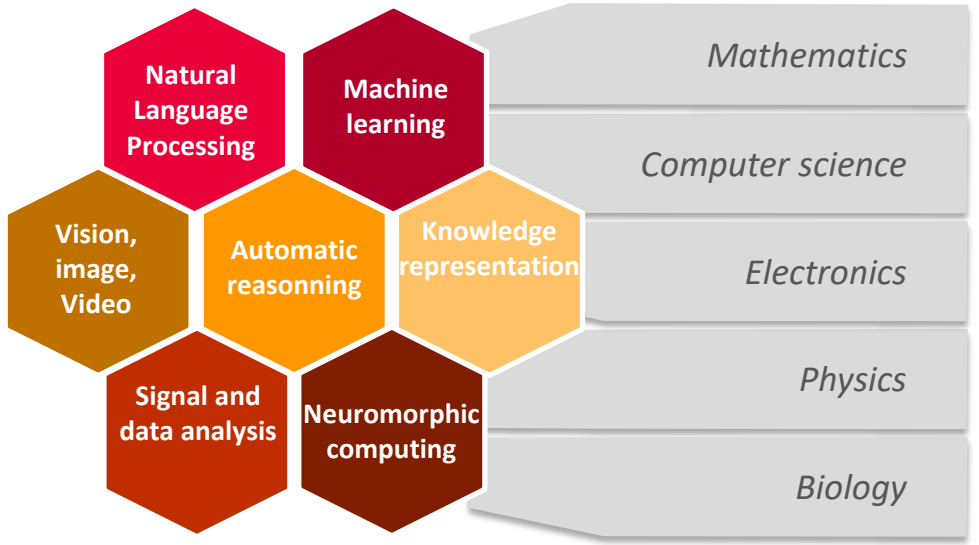
Our research activities, two complementary point of views

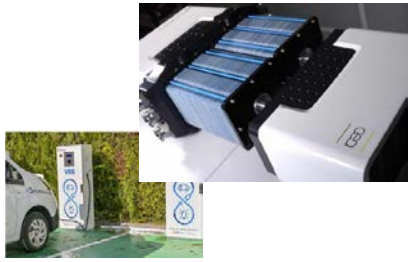
AI for Science and Society

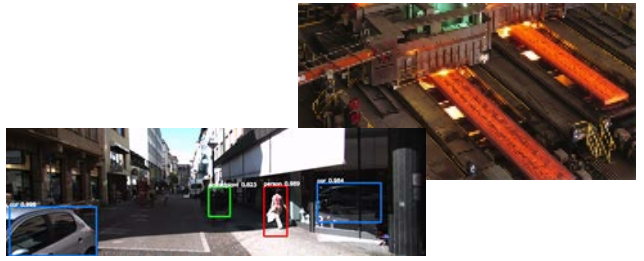




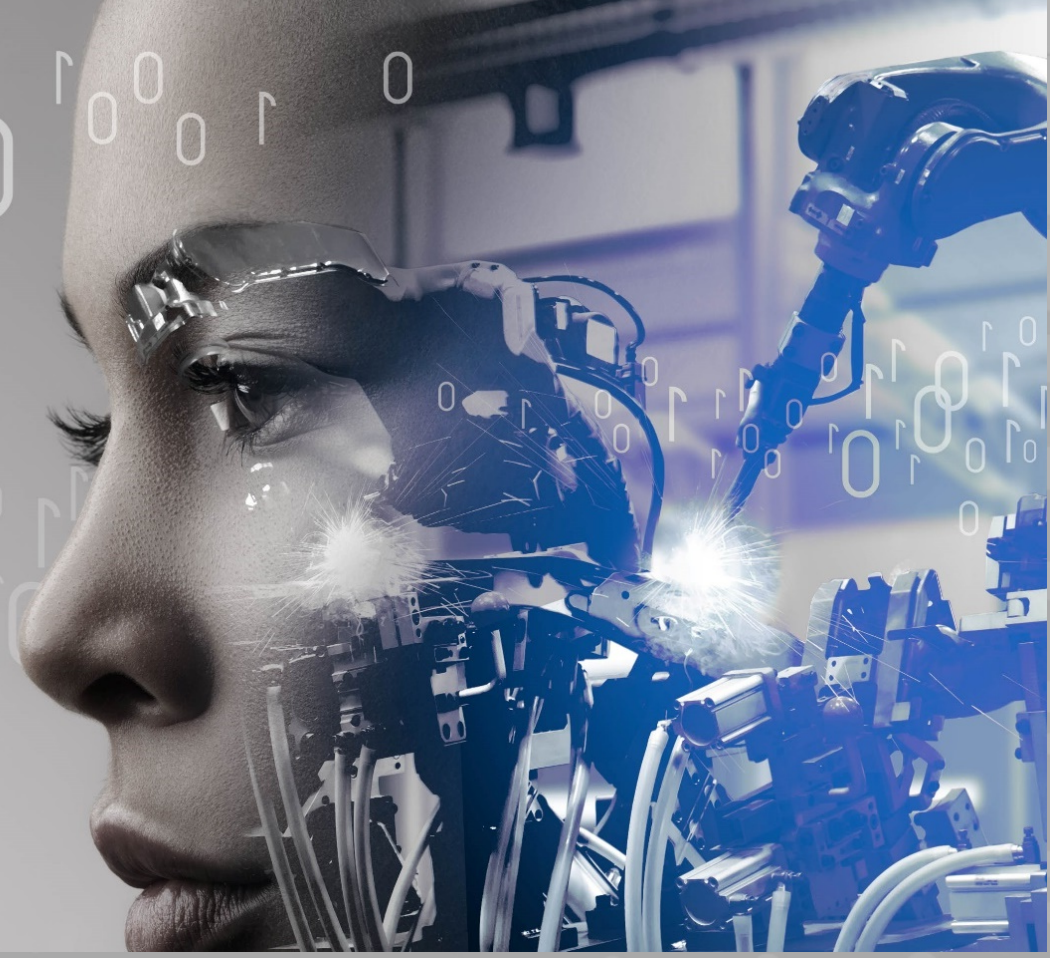
Science for AI







list
cea tech



Expertise and key Programs for digital

Commissariat à l’énergie atomique et aux énergies alternatives - www.cea.fr

OUR EXPERTISE

Software engineering
- Formal mathematical methods
- Model based Soft. & Syst. Eng.

Simulation
- VR & AR, Monte Carlo simulation
- Non-destructive testing

Data intelligence
- Data analytics
- Distrib., embedded & trusted AI

Cyber-security
- Soft. security level assessment
- Network supervision

Robotics
- Collaborative robotics
- Dextral manipulation

Instrumentation & metrology
- Sensors & instrumentation
- IR ref, Radiotherapy & imaging

Computing & embedded syst.
- HPC, quantum, Low pw AI accel.
- Algorithm to chip design

Human-system interfaces
- Haptic interaction
- Surfaces functionalization

An innovative ecosystem

Factory-IA

Computing resource for industrial research

- **170 users, 3 Peta FLOPS, 72x A100, > 300 GPU board**

Clean rooms

Microelectronics technology research requires very large technical infrastructure

…

…

Plan stratégique national de recherche en IA

Accélérer les travaux en IA embarquée, frugale, distribuée, de confiance

➔ Une grande campagne de recrutement de thèses, postdoc, CDD

<u>Des sujets amonts en lien avec l'écosystème académique</u>

- Calcul inspiré de la physique
- Architectures électroniques adaptatives
- Optimisation des modèles de deeplearning
- Méthodes formelles pour la sûreté du ML
- IA décentralisée robuste
- Frugalité en données en en calcul

<u>Des sujets de recherche appliquée en lien avec l'industrie</u>

- Optimisation du déploiement sur les calculateurs
- Certification des systèmes à base d'IA
- Apprentissage embarqué frugal

list
cea tech

Quelques exemples d'application de l'IA

EXAMPLE OF APPLICATION DOMAINS

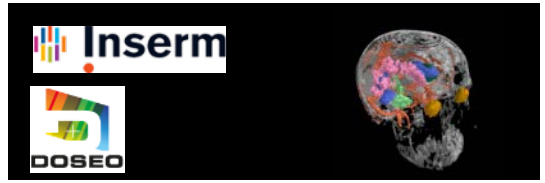
Performance & algorithms for applications

- *Mobility*
- *Energy*
- *Health*
- *Control/diagnosis*
- *Security*
- *Optimization*
- *…*

Vision for autonomous mobility

Natural language processing for data intelligence

Health: diagnosis assistance

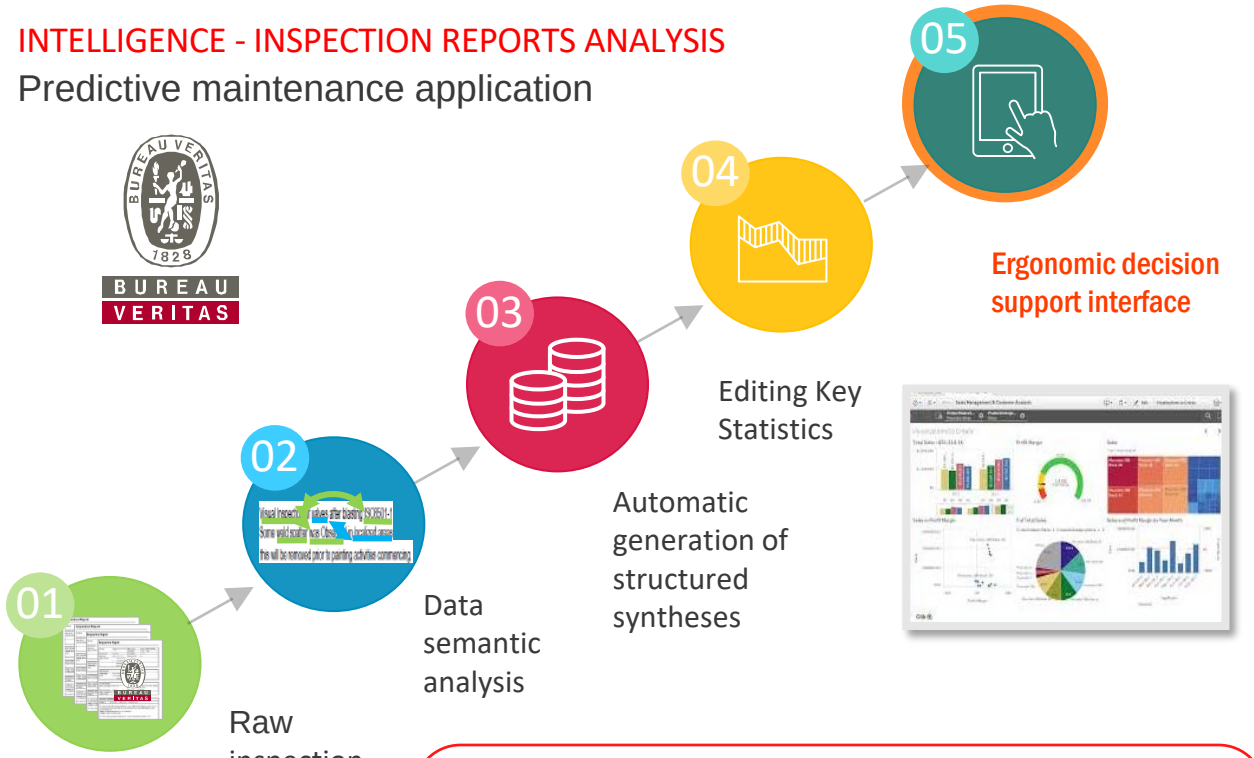


Industry of the future: quality control

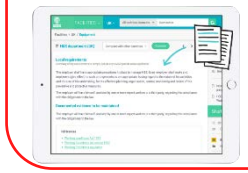
NATURAL LANGUAGE PROCESSING FOR DATA INTELLIGENCE

INTELLIGENCE - INSPECTION REPORTS ANALYSIS

Predictive maintenance application



REGULATORY WATCH PLATFORM

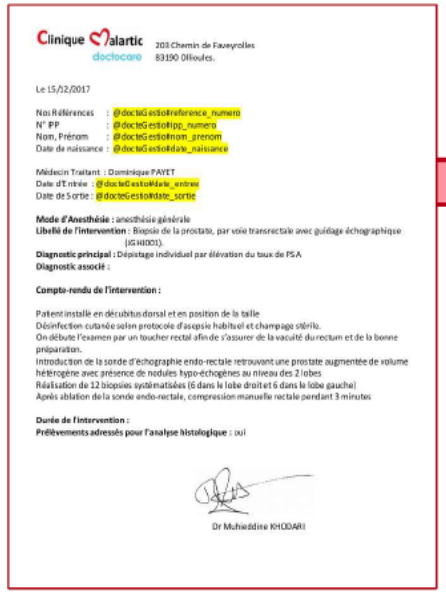


- **12 000** official sources of information
- **20 000** Web pages screened every day
- **>50** countries covered

CLASSIFICATION AUTOMATIQUE : DIAGNOSTICS DES DOSSIERS PATIENTS

DocteGestio

Textes des documents des dossiers patients (comptes-rendus, courriers, ordonnances…)



Clinique Malartic — doctocare — 203 Chemin de Fauveyrolles 83190 Ollioules.

Le 15/12/2017

Noi Références : @docteGestio#reference_numero
N° PP : @docteGestio#pp_numero
Nom, Prénom : @docteGestio#nom_prenom
Date de naissance : @docteGestio#date_naissance

Médecin Traitant : Dominique PAYET
Date d'Entrée : @docteGestio#date_entree
Date de Sortie : @docteGestio#date_sortie

Mode d'Anesthésie : anesthésie générale
Libellé de l'intervention : Biopsie de la prostate, par voie transrectale avec guidage échographique (GSH003).
Diagnostic principal : Dépistage individuel par élévation du taux de PSA
Diagnostic associé :

Compte-rendu de l'intervention :

Patient installé en décubitus dorsal et en position de la taille
Désinfection cutanée selon protocole d'asepsie habituel et champage stérile.
On débute l'examen par un toucher rectal afin de s'assurer de la vacuité du rectum et de la bonne préparation.
Introduction de la sonde d'échographie endo-rectale retrouvant une prostate augmentée de volume hétérogène avec présence de nodules hypo-échogènes au niveau des 2 lobes
Réalisation de 12 biopsies systématisées (6 dans le lobe droit et 6 dans le lobe gauche)
Après ablation de la sonde endo-rectale, compression manuelle rectale pendant 3 minutes

Durée de l'intervention :
Prélèvements adressés pour l'analyse histologique : oui

Dr Muhieddine KHODARI

Classification automatique selon les catégories CIM-10 (*Classification Internationale des maladies*)

Sur les classes avec au moins 200 exemples
P@1 = 0.85
P@3 = 0.93

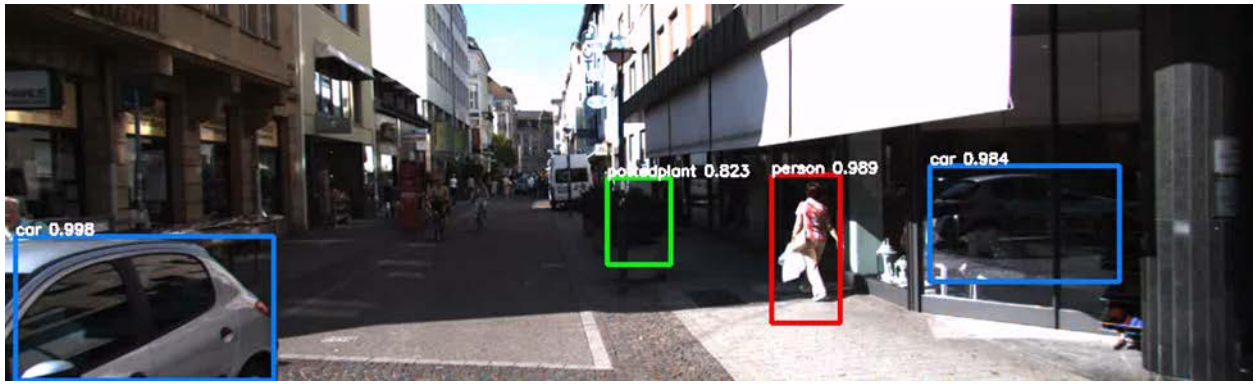
Diagnostic	*score*
C61 Tumeur maligne de la prostate	**0.958984**
N411 Prostatite chronique	*0.0195313*
R778 Autres anomalies précisées des protéines plasmatiques	*0.00976564*

VISION: DEEP LEARNING FOR IMAGE UNDERSTANDING AND PERCEPTION

Some academic BENCHMARK

- **1st rank in vehicle orientation estimation**
- **Top-10 in object detection**

Runs at 10 Hz on Nvidia Gtx 1080



DeepManta

A many-task deep neural network for **visual object recognition**

Detecting Human InterActions By Only Looking Once

- **Top-1 sur Microsoft V-COCO**
- **Top-2 sur HICO-DET data set**

Understanding of environment and of man ⇔ environnement interactions

Geolocation of multi media document

- **1rst MEDIA EVAL 2014-15-16**

Presentation at CES 2018, 2019 – Drive4U from Valeo

Valeo

MAN-ROBOT COLLABORATION

Force capture

Intention detection

Effort control

FRONTIER SCIENCE IN HEATHCARE

Brain controlled exoskeleton

AI FOR MEDICAL DIAGNOSIS ASSISTANCE

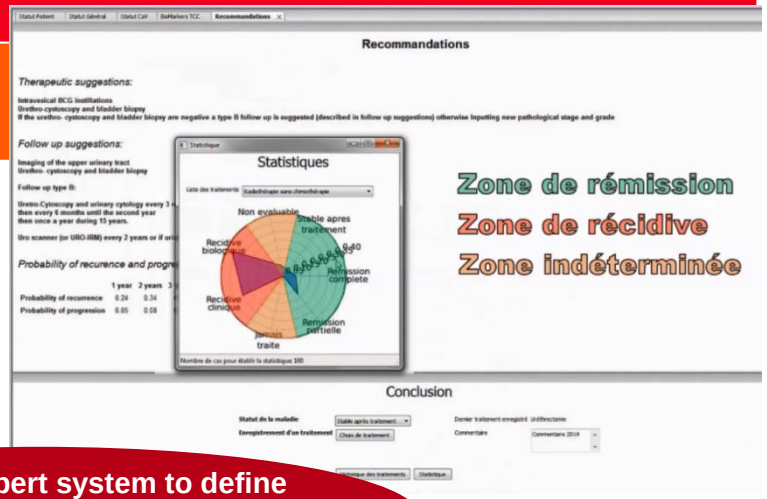
APHP PARTNERSHIP IN AI

ASSISTANCE PUBLIQUE HÔPITAUX DE PARIS

Development of AI application tools in medicine

Olivier CUSSENOT
Urological Surgeon - Paris
TENON HOSPITAL

Multidisciplinary Consultation Meeting (RCP)

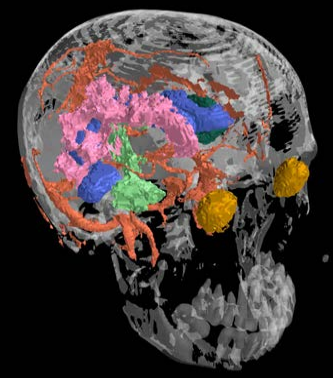


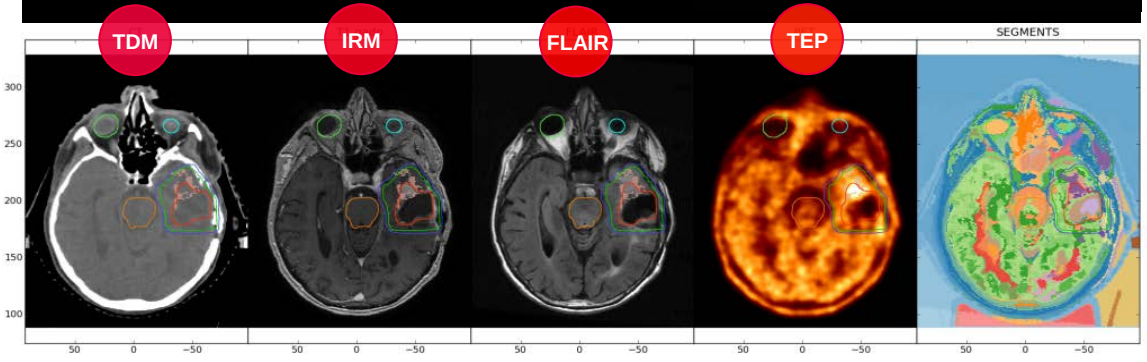
Supervised expert system to define and EXPLAIN the best treatment for each patient

Frédéric Joliot-Curie Hospital Service Partnership

Inserm

DOSEO



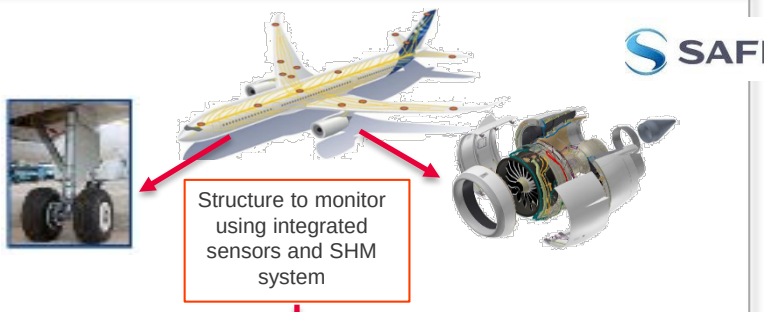


Machine Learning from a database (300 cases) implemented with the Frédéric Joliot-Curie Hospital Service (SHFJ) of Orsay, allowing:

- Identification of tissues, especially tumors
- 3D visualization of vascularization
- Identification of new pre-figurative markers of future developments

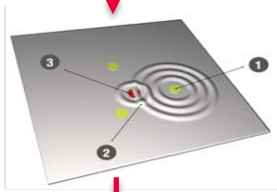
MANUFACTURING: PRODUCTION CONTROL AND OPTIMISATION

SAFRAN

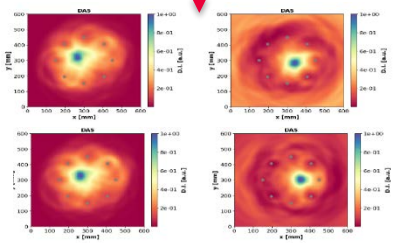


Structure to monitor using integrated sensors and SHM system

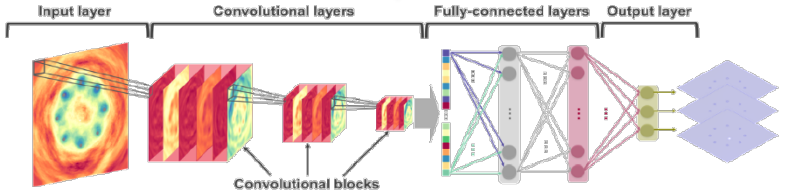
GWs-SHM schematic:
(1) Emiter/Receptor
(2) Diffracted wawes
(3) Defect



Guided elastic wave inspection for damage monitoring



Data for different defect sizes generated with CIVA

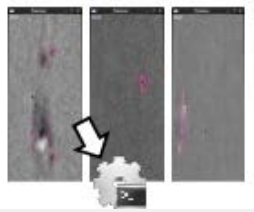


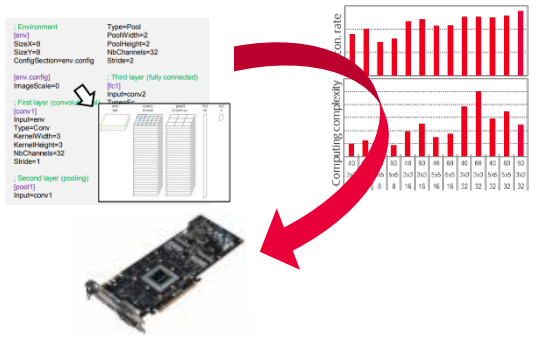
ArcelorMittal

Automatized quality control of steel during the rolling process

- High speed real time control (20 m/s)
- Loock for small defaults (~mm) with low constrast
- Hard environnement (oil vapors, smallness, etc.)

N2D2





- Image analysis: several Mpixels in few seconds
- Default detection rate: 95 %

list
cea tech

Plus d’information sur le site web : https://list.cea.fr/